

```
In [2]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
In [3]: df = pd.read_csv(r"C:\Users\User\Desktop\dv.csv")
df
```

```
Out[3]:
```

|  | name | planet_status | mass | mass_error_min | mass_error_max | mass_sini | mass_sini |
|--|------|---------------|------|----------------|----------------|-----------|-----------|
|--|------|---------------|------|----------------|----------------|-----------|-----------|

|          |              |           |       |         |         |         |  |
|----------|--------------|-----------|-------|---------|---------|---------|--|
| <b>0</b> | 109<br>Psc b | Confirmed | 5.743 | 0.28900 | 1.01100 | 6.38300 |  |
|----------|--------------|-----------|-------|---------|---------|---------|--|

|          |              |           |     |         |         |         |  |
|----------|--------------|-----------|-----|---------|---------|---------|--|
| <b>1</b> | 112<br>Psc b | Confirmed | NaN | 0.00500 | 0.00400 | 0.03300 |  |
|----------|--------------|-----------|-----|---------|---------|---------|--|

|          |              |           |       |         |         |     |  |
|----------|--------------|-----------|-------|---------|---------|-----|--|
| <b>2</b> | 112<br>Psc c | Confirmed | 9.866 | 1.78100 | 3.19000 | NaN |  |
|----------|--------------|-----------|-------|---------|---------|-----|--|

|          |                |           |     |         |         |          |  |
|----------|----------------|-----------|-----|---------|---------|----------|--|
| <b>3</b> | 11<br>UMi<br>b | Confirmed | NaN | 1.10000 | 1.10000 | 11.08730 |  |
|----------|----------------|-----------|-----|---------|---------|----------|--|

|          |                 |           |     |         |         |         |  |
|----------|-----------------|-----------|-----|---------|---------|---------|--|
| <b>4</b> | 14<br>And<br>Ab | Confirmed | NaN | 0.23000 | 0.23000 | 4.68400 |  |
|----------|-----------------|-----------|-----|---------|---------|---------|--|

|     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|
| ... | ... | ... | ... | ... | ... | ... | ... |
|-----|-----|-----|-----|-----|-----|-----|-----|

|             |              |           |     |         |         |         |  |
|-------------|--------------|-----------|-----|---------|---------|---------|--|
| <b>5981</b> | YBP<br>401 b | Confirmed | NaN | 0.05000 | 0.05000 | 0.46000 |  |
|-------------|--------------|-----------|-----|---------|---------|---------|--|

|             |             |           |       |         |         |     |  |
|-------------|-------------|-----------|-------|---------|---------|-----|--|
| <b>5982</b> | YSES<br>2 b | Confirmed | 6.300 | 0.90000 | 1.60000 | NaN |  |
|-------------|-------------|-----------|-------|---------|---------|-----|--|

|             |             |           |     |         |         |         |  |
|-------------|-------------|-----------|-----|---------|---------|---------|--|
| <b>5983</b> | YZ<br>Cet b | Confirmed | NaN | 0.00028 | 0.00028 | 0.00220 |  |
|-------------|-------------|-----------|-----|---------|---------|---------|--|

|             |             |           |     |         |         |         |  |
|-------------|-------------|-----------|-----|---------|---------|---------|--|
| <b>5984</b> | YZ<br>Cet c | Confirmed | NaN | 0.00035 | 0.00035 | 0.00359 |  |
|-------------|-------------|-----------|-----|---------|---------|---------|--|

|             |             |           |     |         |         |         |  |
|-------------|-------------|-----------|-----|---------|---------|---------|--|
| <b>5985</b> | YZ<br>Cet d | Confirmed | NaN | 0.00038 | 0.00038 | 0.00343 |  |
|-------------|-------------|-----------|-----|---------|---------|---------|--|

5986 rows × 10 columns



```
In [4]: print("Shape:", df.shape)
```

Shape: (5986, 10)

```
In [5]: df.fillna("Nil")
```

Out[5]:

|             | name            | planet_status | mass  | mass_error_min | mass_error_max | mass_sini | mass_sini |
|-------------|-----------------|---------------|-------|----------------|----------------|-----------|-----------|
| <b>0</b>    | 109<br>Psc b    | Confirmed     | 5.743 | 0.289          | 1.011          | 6.383     |           |
| <b>1</b>    | 112<br>Psc b    | Confirmed     | Nil   | 0.005          | 0.004          | 0.033     |           |
| <b>2</b>    | 112<br>Psc c    | Confirmed     | 9.866 | 1.781          | 3.19           | Nil       |           |
| <b>3</b>    | 11<br>UMi<br>b  | Confirmed     | Nil   | 1.1            | 1.1            | 11.0873   |           |
| <b>4</b>    | 14<br>And<br>Ab | Confirmed     | Nil   | 0.23           | 0.23           | 4.684     |           |
| ...         | ...             | ...           | ...   | ...            | ...            | ...       | ...       |
| <b>5981</b> | YBP<br>401 b    | Confirmed     | Nil   | 0.05           | 0.05           | 0.46      |           |
| <b>5982</b> | YSES<br>2 b     | Confirmed     | 6.3   | 0.9            | 1.6            | Nil       |           |
| <b>5983</b> | YZ<br>Cet b     | Confirmed     | Nil   | 0.00028        | 0.00028        | 0.0022    |           |
| <b>5984</b> | YZ<br>Cet c     | Confirmed     | Nil   | 0.00035        | 0.00035        | 0.00359   |           |
| <b>5985</b> | YZ<br>Cet d     | Confirmed     | Nil   | 0.00038        | 0.00038        | 0.00343   |           |

5986 rows × 10 columns



```
In [6]: print("\nColumns and Data Types:")
        print(df.info())
```

Columns and Data Types:

&lt;class 'pandas.core.frame.DataFrame'&gt;

RangeIndex: 5986 entries, 0 to 5985

Data columns (total 10 columns):

| # | Column              | Non-Null Count | Dtype   |
|---|---------------------|----------------|---------|
| 0 | name                | 5986 non-null  | object  |
| 1 | planet_status       | 5986 non-null  | object  |
| 2 | mass                | 2079 non-null  | float64 |
| 3 | mass_error_min      | 2817 non-null  | float64 |
| 4 | mass_error_max      | 2817 non-null  | float64 |
| 5 | mass_sini           | 1202 non-null  | float64 |
| 6 | mass_sini_error_min | 1039 non-null  | float64 |
| 7 | mass_sini_error_max | 1039 non-null  | float64 |
| 8 | radius              | 4568 non-null  | float64 |
| 9 | radius_error_min    | 4093 non-null  | float64 |

dtypes: float64(8), object(2)

memory usage: 467.8+ KB

None

```
In [7]: print("\nFirst 5 rows:")
        print(df.head())
```

First 5 rows:

|   | name      | planet_status | mass  | mass_error_min | mass_error_max | mass_sini | \ |
|---|-----------|---------------|-------|----------------|----------------|-----------|---|
| 0 | 109 Psc b | Confirmed     | 5.743 | 0.289          | 1.011          | 6.3830    |   |
| 1 | 112 Psc b | Confirmed     | NaN   | 0.005          | 0.004          | 0.0330    |   |
| 2 | 112 Psc c | Confirmed     | 9.866 | 1.781          | 3.190          | NaN       |   |
| 3 | 11 UMi b  | Confirmed     | NaN   | 1.100          | 1.100          | 11.0873   |   |
| 4 | 14 And Ab | Confirmed     | NaN   | 0.230          | 0.230          | 4.6840    |   |

|   | mass_sini_error_min | mass_sini_error_max | radius | radius_error_min |
|---|---------------------|---------------------|--------|------------------|
| 0 | 0.078               | 0.078               | 1.152  | NaN              |
| 1 | 0.005               | 0.004               | NaN    | NaN              |
| 2 | NaN                 | NaN                 | NaN    | NaN              |
| 3 | 1.100               | 1.100               | NaN    | NaN              |
| 4 | 0.230               | 0.230               | NaN    | NaN              |

```
In [8]: print("Planet Status counts:")
        print(df['planet_status'].value_counts())
        print("\nUnique names or duplicates:")
        print("Duplicate rows:", df.duplicated().sum())
        print("Duplicate names:", df['name'].duplicated().sum())
        print("\nMissing values summary:")
        print(df.isnull().sum())
```

Planet Status counts:

```
planet_status
Confirmed    5986
Name: count, dtype: int64
```

Unique names or duplicates:

```
Duplicate rows: 0
Duplicate names: 0
```

Missing values summary:

```
name          0
planet_status  0
mass          3907
mass_error_min 3169
mass_error_max 3169
mass_sini      4784
mass_sini_error_min 4947
mass_sini_error_max 4947
radius        1418
radius_error_min 1893
dtype: int64
```

```
In [9]: print("Numeric Descriptive Statistics:")
        print(df.describe())
```

## Numeric Descriptive Statistics:

|       | mass        | mass_error_min | mass_error_max | mass_sini   | \ |
|-------|-------------|----------------|----------------|-------------|---|
| count | 2079.000000 | 2817.000000    | 2817.0000      | 1202.000000 |   |
| mean  | 1.492538    | 0.324366       | inf            | 1.983529    |   |
| std   | 2.602286    | 1.351929       | NaN            | 3.199717    |   |
| min   | 0.000190    | 0.000000       | -17.8000       | 0.000470    |   |
| 25%   | 0.029655    | 0.006290       | 0.0057         | 0.045000    |   |
| 50%   | 0.370000    | 0.041000       | 0.0400         | 0.731500    |   |
| 75%   | 1.466500    | 0.170000       | 0.1700         | 2.400000    |   |
| max   | 12.934000   | 53.500000      | inf            | 50.000000   |   |

|       | mass_sini_error_min | mass_sini_error_max | radius      | radius_error_min |
|-------|---------------------|---------------------|-------------|------------------|
| count | 1039.000000         | 1039.000000         | 4568.000000 | 4093.000000      |
| mean  | 0.214687            | 0.240453            | 0.401515    | 0.048986         |
| std   | 0.636474            | 0.945468            | 0.430601    | 0.754220         |
| min   | 0.000000            | 0.000000            | 0.025430    | 0.000000         |
| 25%   | 0.005940            | 0.005300            | 0.145000    | 0.010000         |
| 50%   | 0.049300            | 0.050000            | 0.219000    | 0.020000         |
| 75%   | 0.170000            | 0.170000            | 0.384000    | 0.042000         |
| max   | 13.000000           | 20.900000           | 6.900000    | 48.000000        |

```
In [10]: print("Infinite values in mass_error_max:", np.isinf(df['mass_error_max']).sum())
print("Negative values in mass_error_max:", (df['mass_error_max'] < 0).sum())
print("Rows with negative mass_error_max:")
print(df[df['mass_error_max'] < 0])
print("\nRows with inf mass_error_max:")
print(df[np.isinf(df['mass_error_max'])])
```

Infinite values in mass\_error\_max: 9

Negative values in mass\_error\_max: 1

Rows with negative mass\_error\_max:

|     | name       | planet_status | mass | mass_error_min | mass_error_max \ |
|-----|------------|---------------|------|----------------|------------------|
| 803 | HD 19615 b | Confirmed     | 8.6  | 53.5           | -17.8            |

|     | mass_sini | mass_sini_error_min | mass_sini_error_max | radius \ |
|-----|-----------|---------------------|---------------------|----------|
| 803 | 8.5       | 0.7                 | 0.7                 | NaN      |

|     | radius_error_min |
|-----|------------------|
| 803 | NaN              |

Rows with inf mass\_error\_max:

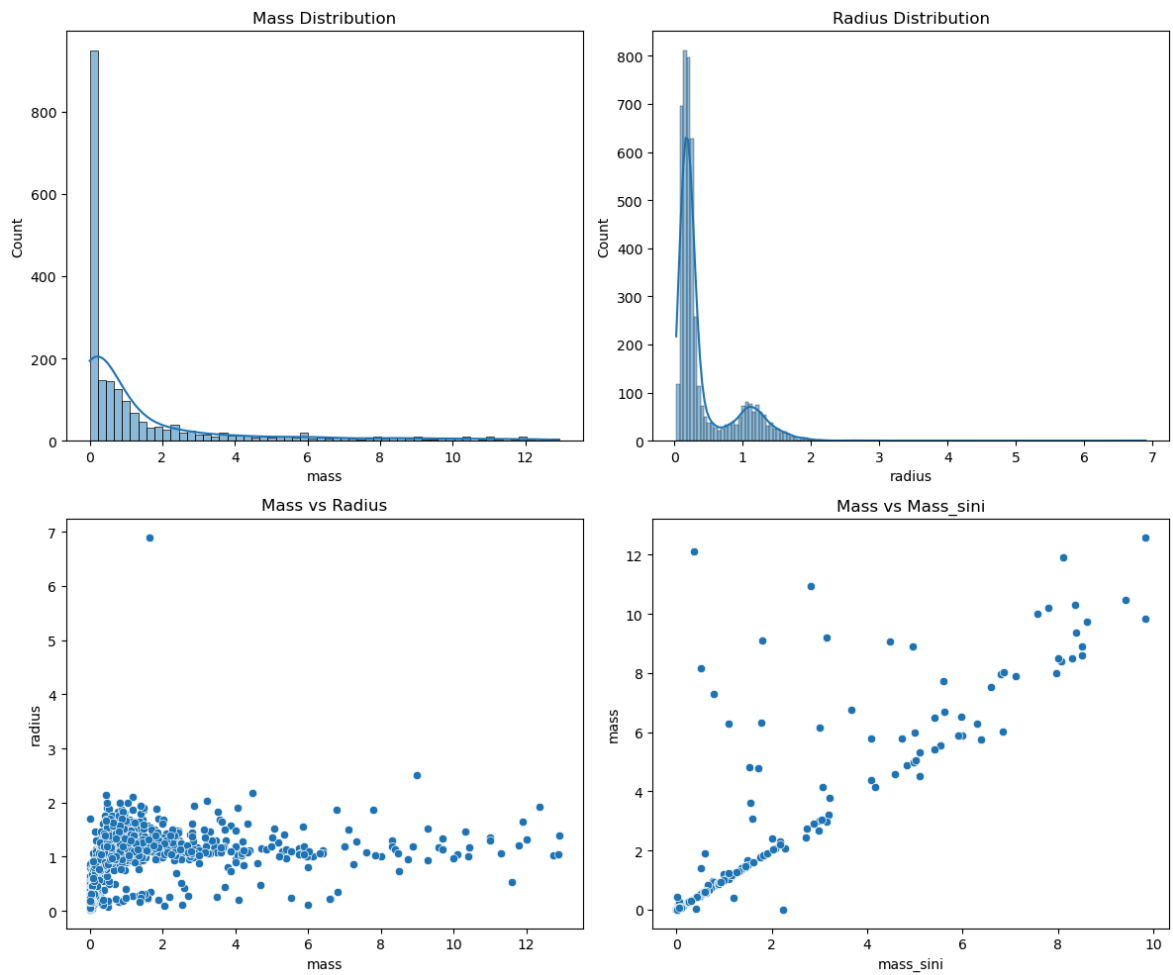
|      | name        | planet_status | mass     | mass_error_min | mass_error_max \ |
|------|-------------|---------------|----------|----------------|------------------|
| 92   | CD Cet b    | Confirmed     | 0.01244  | 0.00136        | inf              |
| 768  | HD 181720 b | Confirmed     | 12.12000 | 5.88000        | inf              |
| 1093 | HD 69830 b  | Confirmed     | 0.14270  | 0.09280        | inf              |
| 1094 | HD 69830 c  | Confirmed     | 0.16490  | 0.10720        | inf              |
| 1095 | HD 69830 d  | Confirmed     | 0.25290  | 0.16440        | inf              |
| 5238 | TOI-1416 c  | Confirmed     | 0.06800  | 0.00880        | inf              |
| 5529 | TOI-431 c   | Confirmed     | 0.00892  | 0.00107        | inf              |
| 5701 | TOI-784 c   | Confirmed     | 0.06878  | 0.02912        | inf              |
| 5749 | ups And Ab  | Confirmed     | 0.62000  | 0.09000        | inf              |

|      | mass_sini | mass_sini_error_min | mass_sini_error_max | radius \ |
|------|-----------|---------------------|---------------------|----------|
| 92   | 0.01243   | 0.00135             | 0.00135             | NaN      |
| 768  | 0.37000   | NaN                 | NaN                 | NaN      |
| 1093 | 0.03210   | NaN                 | NaN                 | NaN      |
| 1094 | 0.03710   | NaN                 | NaN                 | NaN      |
| 1095 | 0.05690   | NaN                 | NaN                 | NaN      |
| 5238 | 0.06800   | 0.00880             | 0.00980             | NaN      |
| 5529 | 0.00890   | 0.00107             | 0.00129             | NaN      |
| 5701 | 0.06875   | 0.02910             | 0.02910             | NaN      |
| 5749 | 0.62000   | 0.09000             | 0.09000             | NaN      |

|      | radius_error_min |
|------|------------------|
| 92   | NaN              |
| 768  | NaN              |
| 1093 | NaN              |
| 1094 | NaN              |
| 1095 | NaN              |
| 5238 | NaN              |
| 5529 | NaN              |
| 5701 | NaN              |
| 5749 | NaN              |

```
In [11]: fig, axes = plt.subplots(2, 2, figsize=(12, 10))
sns.histplot(df['mass'].dropna(), ax=axes[0,0], kde=True).set_title('Mass Distri
sns.histplot(df['radius'].dropna(), ax=axes[0,1], kde=True).set_title('Radius Di
sns.scatterplot(data=df, x='mass', y='radius', ax=axes[1,0]).set_title('Mass vs
sns.scatterplot(data=df, x='mass_sini', y='mass', ax=axes[1,1]).set_title('Mass
plt.tight_layout()
plt.savefig('distributions.png')
print("Plots saved.")
```

Plots saved.



```
In [12]: corr = df[['mass', 'mass_sini', 'radius']].corr()
print("Correlation Matrix:")
print(corr)
```

Correlation Matrix:

|           | mass     | mass_sini | radius   |
|-----------|----------|-----------|----------|
| mass      | 1.000000 | 0.875983  | 0.417286 |
| mass_sini | 0.875983 | 1.000000  | 0.130681 |
| radius    | 0.417286 | 0.130681  | 1.000000 |

In [ ]: